Code of the student

☐0	☐0	☐0	☐0	☐0	☐0	☐0	☐0
☐1	☐1	☐1	☐1	☐1	☐1	☐1	☐1
☐2	☐2	☐2	☐2	☐2	☐2	☐2	☐2
☐3	☐3	☐3	☐3	☐3	☐3	☐3	☐3
☐4	☐4	☐4	☐4	☐4	☐4	☐4	☐4
☐5	☐5	☐5	☐5	☐5	☐5	☐5	☐5
☐6	☐6	☐6	☐6	☐6	☐6	☐6	☐6
☐7	☐7	☐7	☐7	☐7	☐7	☐7	☐7
☐8	☐8	☐8	☐8	☐8	☐8	☐8	☐8
☐9	☐9	☐9	☐9	☐9	☐9	☐9	☐9

Surname, Name:

.....

OVNET EXAM (21st of January 2019)

IMPORTANT REMARKS:

- It is important to fill the ID number of the student and the code of the test.
- IT IS COMPLETELY FORBIDDEN TO USE ANY KIND OF COMMUNICATIONS DEVICE.

Pregunta 1 According to RFC 4566 about SDP, which of these statements is FALSE regarding the Glossary?

☐ Session Announcement: the set of protocols, senders and receivers needed to properly establish, maintain and finish multimedia sessions.

☐ Conference: A multimedia conference is a set of two or more communicating users along with the software they are using to communicate.

☐ Session: A multimedia session is a set of multimedia senders and receivers and the data streams flowing from senders to receivers.

☐ Session Description: A well-defined format for conveying sufficient information to discover and participate in a multimedia session.

Pregunta 2 We use the scenario of Figure 2, that also corresponds to the **sip-2proxies** lab session. Both **alice** and **bob** are properly registered in their proxies. You establish and end a successful call from **alice** to **bob**, using the SIP trapezoid. In the `INVITE` request that **prox1** sends to **prox2**, according to the wireshark traces in **SimNet0**, it is FALSE that:

☐ There are two `Via` headers, so the responses will traverse first **prox1** and finally **alice**.

☐ The `Contact` header contains the way to contact **prox1**, that is, `sip:alice@203.0.113.2:5060`

☐ The `To` and **From** headers contain the AoRs of **bob** and **alice** respectively

☐ **prox1** has included a `Record-Route` header, that did not exist in the original `INVITE` sent by **alice**.

Pregunta 3 We use the scenario of Figure 2, that also corresponds to the **sip-2proxies** lab session. You start the registration process of **alice** by executing:

```
alice$ pjsua --id sip:alice@example.org
--registrar sip:example.org --realm
example.org --username alice --password
xxxx --null-audio --auto-play --play-file
hello16000.wav --auto-answer 180 --outbound
sip:example.org --add-codec=PCMA/8000
--nameserver=ns1.example.org
```

The corresponding `kamailio` server is properly configured in **prox1**. According to the SIP dialoque, it is FALSE that:

☐ **prox1** responds to the first `REGISTER` with a 401 `Unauthorized`.

☐ Four SIP packets are needed to register **alice** in the kamailio server **prox1**.

☐ The first `REGISTER` request contains a `WWW-Authenticate` header with a `nonce`, so **prox1** can demostrate to **alice** that it knows the password.

☐ **alice** sends an initial SIP request `REGISTER`, with the `To` and `From` headers containing the AOR of this user.

Pregunta 4 We use the scenario of Figure 2, that also corresponds to the **sip-2proxies** lab session. You start the registration process of **alice** by executing:

```
alice$ pjsua --id sip:alice@example.org
--registrar sip:example.org --realm
example.org --username alice --password
xxxx --null-audio --auto-play --play-file
hello16000.wav --auto-answer 180 --outbound
sip:example.org --add-codec=PCMA/8000
--nameserver=ns1.example.org
```

The corresponding `kamailio` server is properly configured in **prox1**. According to the initial DNS dialoque, it is TRUE that:

☐ The DNS does not need to send any A record, as all the information needed is contained in the SRV record.

☐ **alice** sends a DNS request to know the NAPTR record corresponding to the realm `example.com`.

☐ **router** is sending a DNS request to know the SRV record corresponding to the `sip` service, the realm `example.org` and the transport protocol `TCP`.

☐ **alice** is sending a DNS request to know the SRV record corresponding to the `sip` service, the realm `example.org` and the transport protocol `UDP`.

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Pregunta 5 We use the same scenario of Figure 1, that also corresponds to the **sip-basic** lab session. You establish and end a successful call from **alice** to **bob** using `pjsua` using these commands:

```
alice: # pjsua --null-audio --local-port=5060
bob: # pjsua --play-file /root/hello16000.wav
--auto-answer 180 --auto-play --local-port=5000
--null-audio
```

If you have a look to the `branch` field in the SIP packets, it is FALSE that:

- ☐ The `branch` field is included in the `Via` header
- ☐ The `branch` field is used to identify a SIP transaction
- ☐ In the first transaction, `branch` is the same value in all messages of the three way handshake (`INVITE-OK-ACK`)
- ☐ The `branch` field is the same value for the `BYE-OK` messages that end the call

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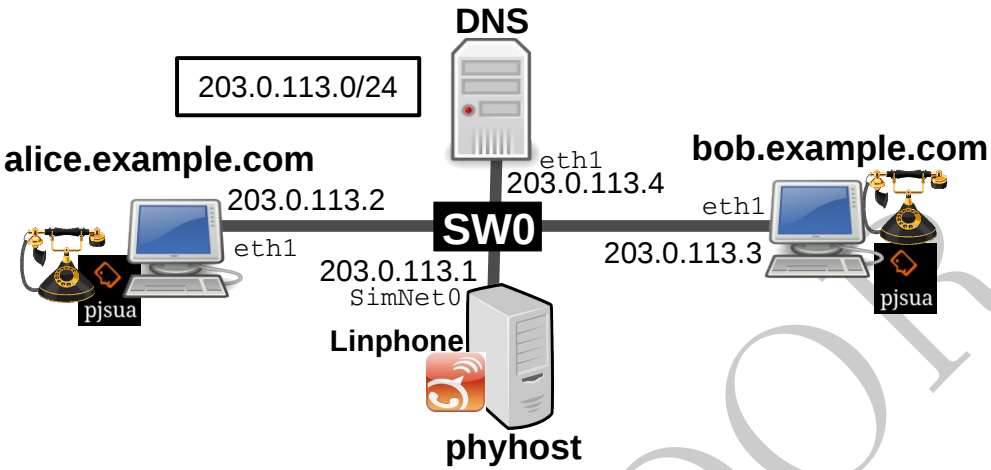


Fig. 1: SIP basic scenario

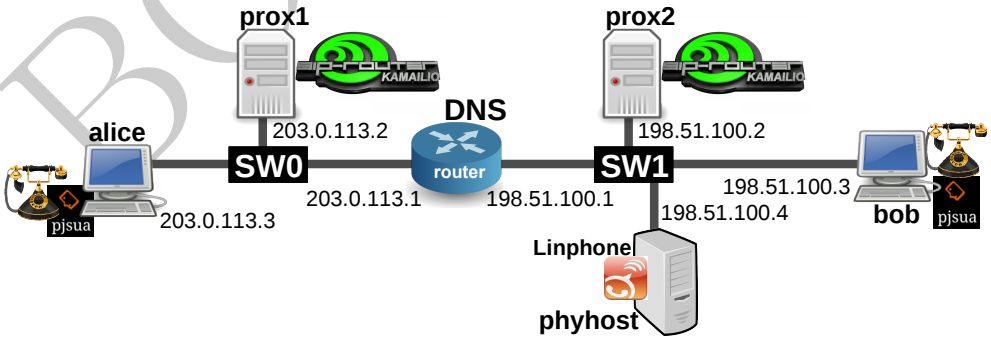


Fig. 2: SIP 2-proxies scenario

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